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u. penn masters of architecture '28

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contents + credits

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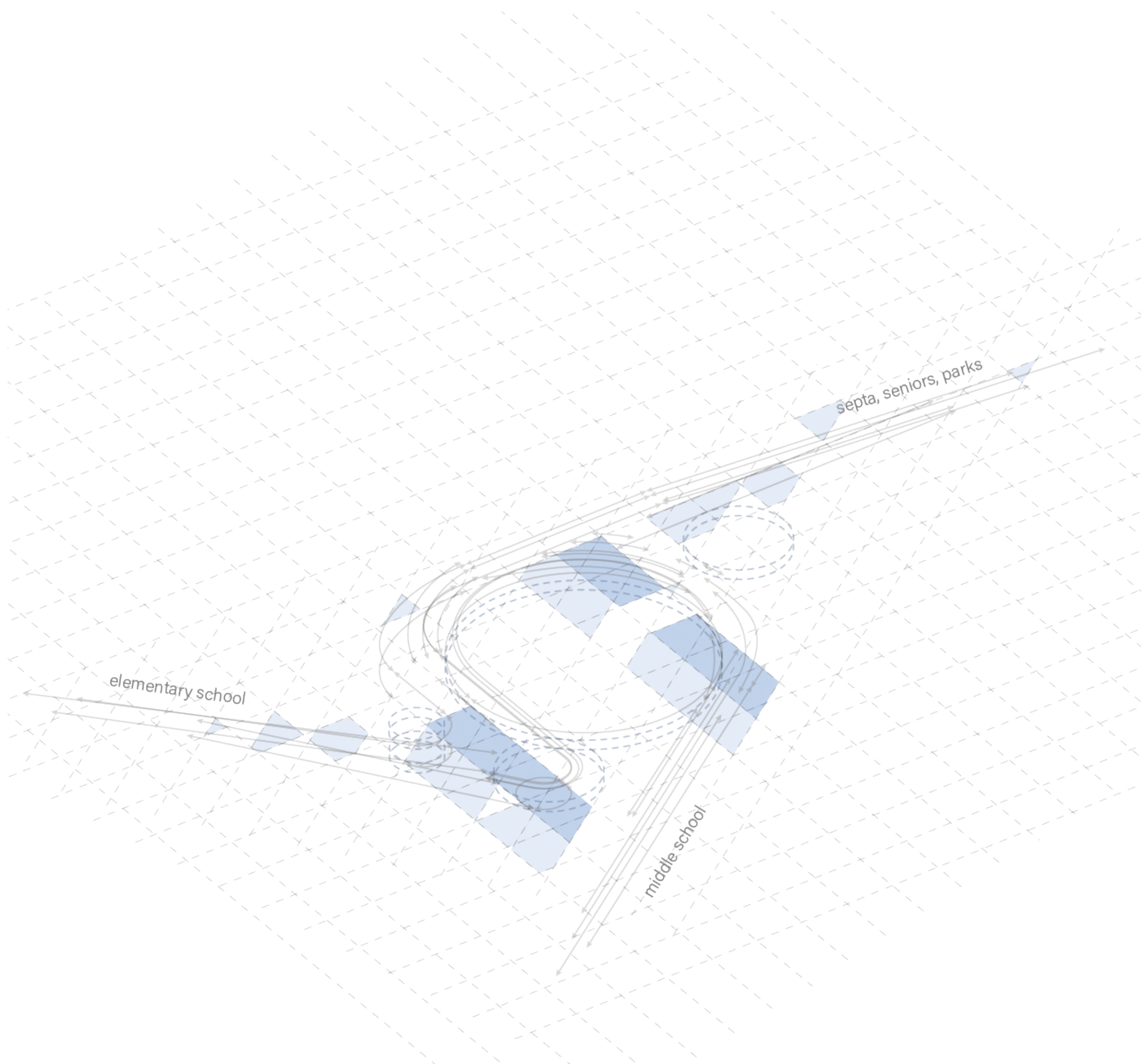
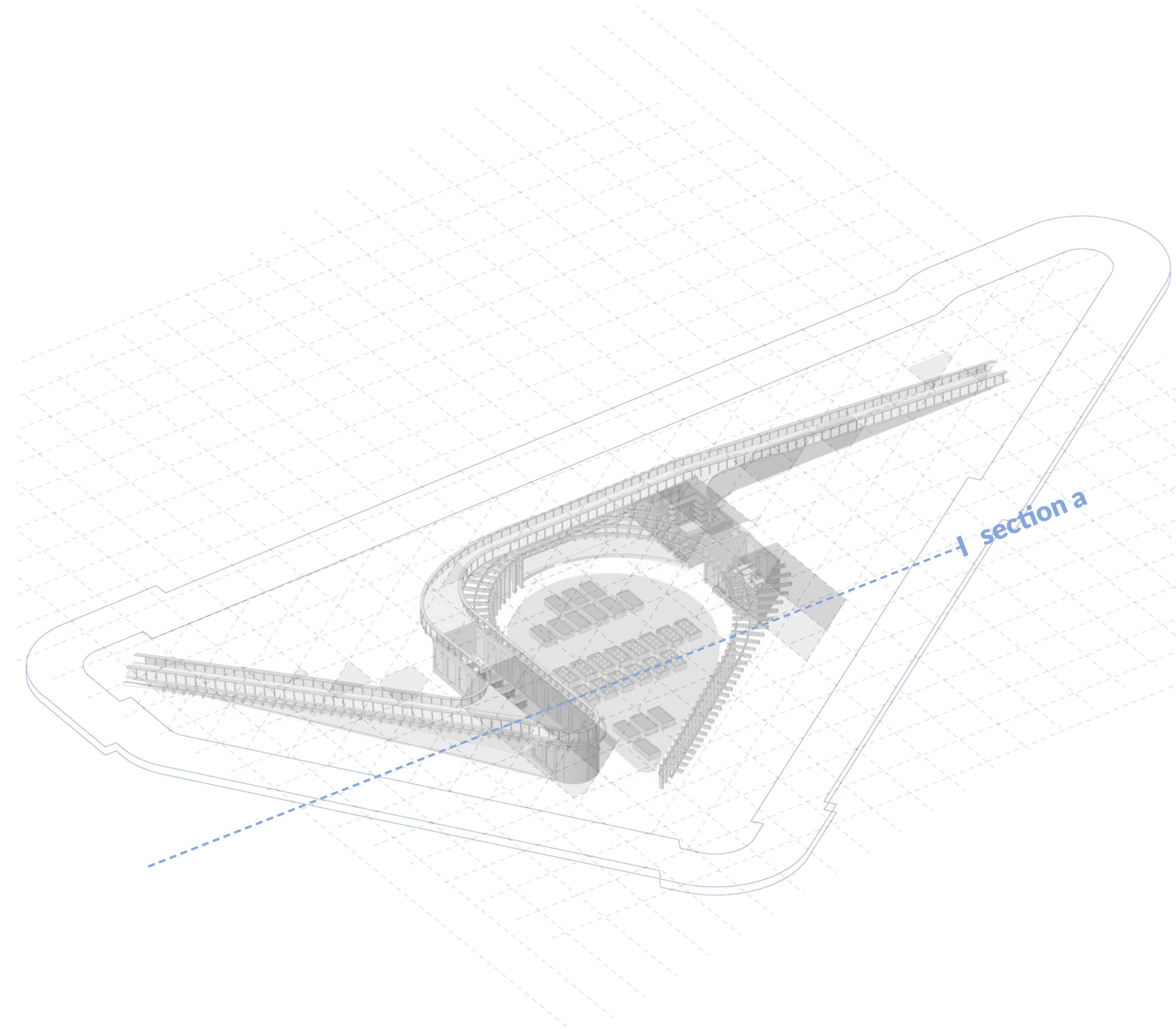
concept collage/diagram:
passing down a recipe

food and nature bring together generations: every child loves their grandmother's recipes and the smell of freshly picked herbs.

Blanche A. Nixon Library
project site + programming

this pavilion is an intervention on the site of a library in Cobb's Creek, in West Philadelphia. The librarian requested our program include both senior and youth programming and so we combined the two through food and nature: gardening, and cooking classes..



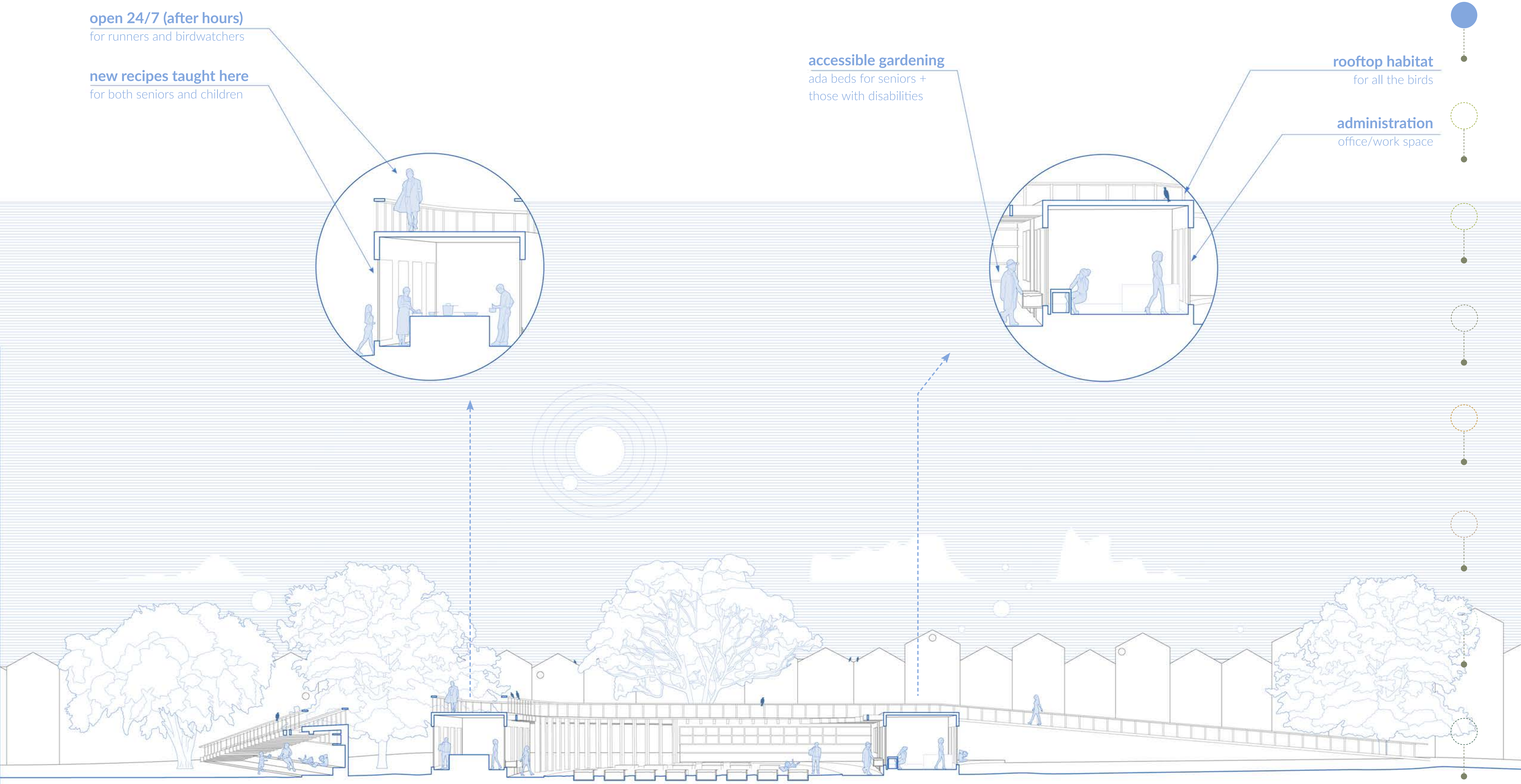


form-finding diagram:
the site + its occupants

the four blocks surrounding the project site have an elementary school, a middle school, a transit stop, and parks. Our form is derived from their circulation.

site axonometric diagram:
flowing around a garden

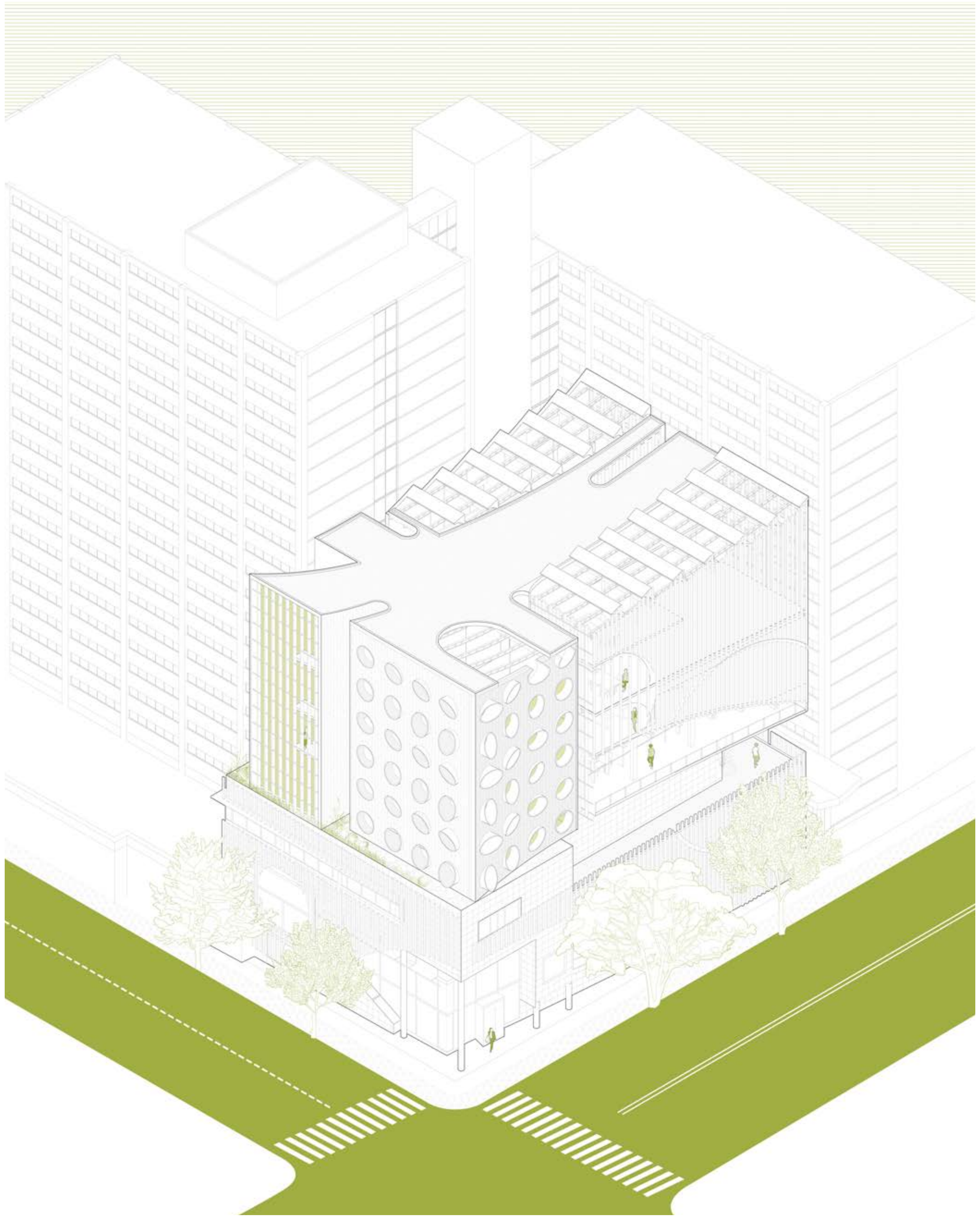
the competition prompt required that the enclosed structure have very little roof area but still fully secure the site. we used ramps to create this enclosure,

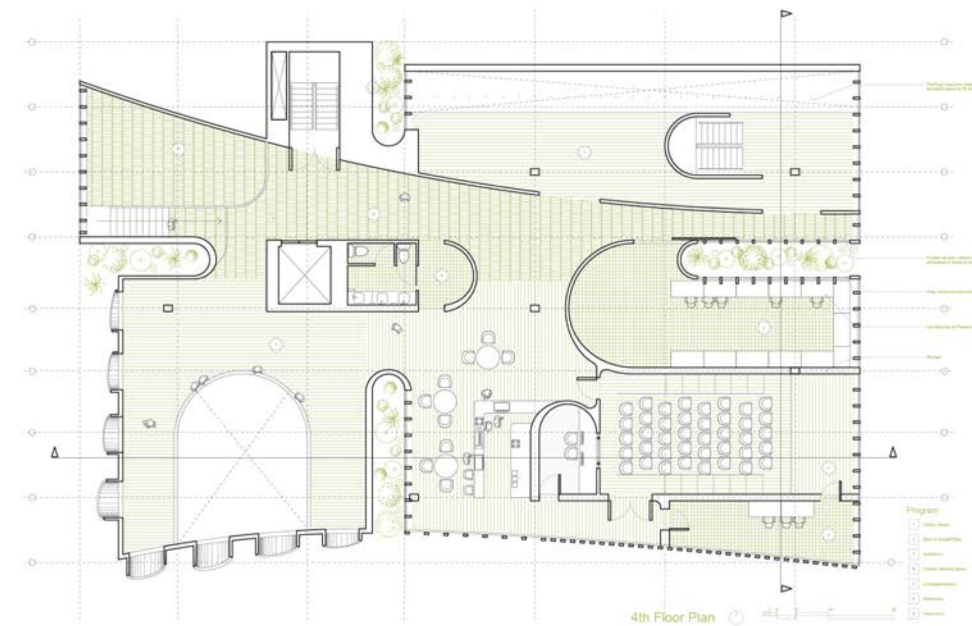


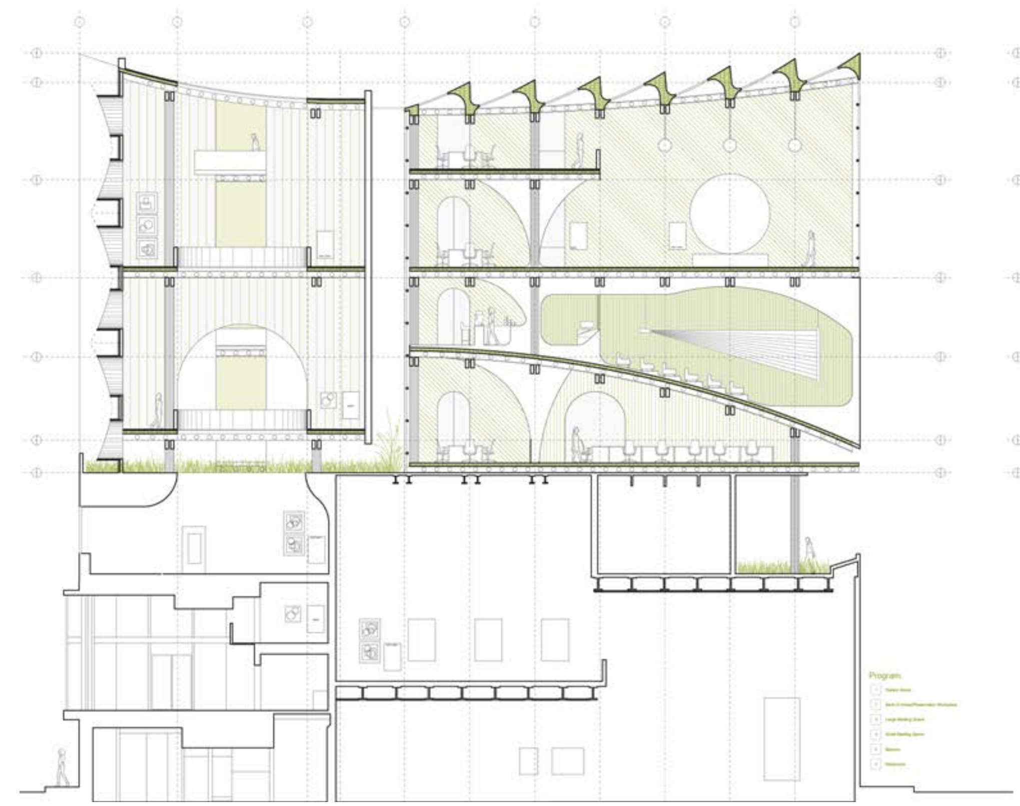
section a

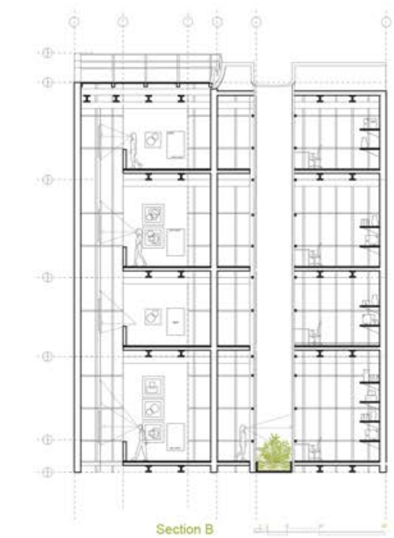
recipe

competition, spring 2026









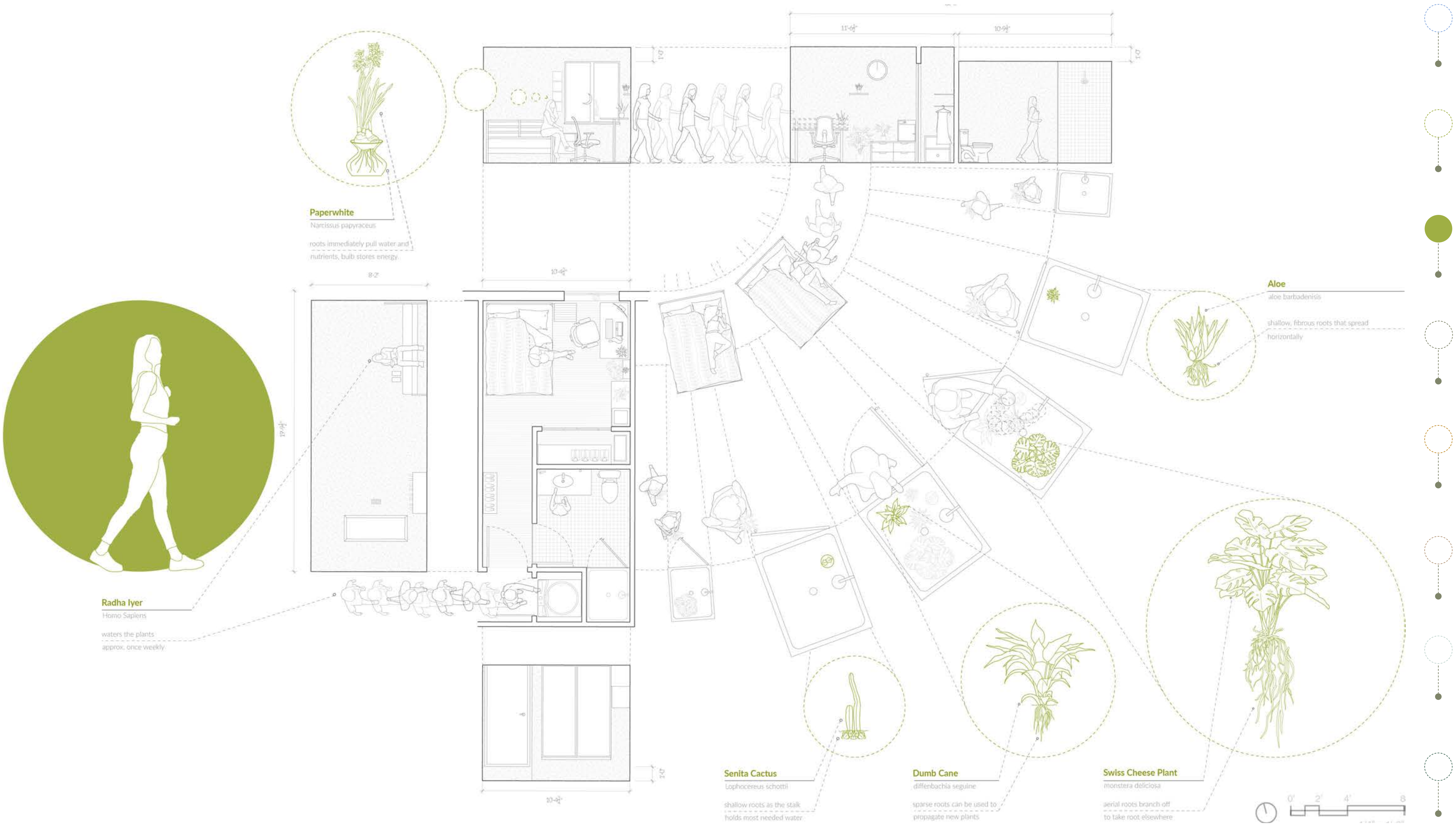
cutting space with light:

a vertical extension to the institute of contemporary art. In plan and section, 5' circles divide space, 10' circles create smaller private spaces, 20' circles allow in light or create shared public spaces, shaping movement with light. Program includes back-of-house space for art preservationists, a multipurpose gallery/event space, and a small auditorium.

ica extension: a vertical extension for the institute of contemporary art, Philadelphia.

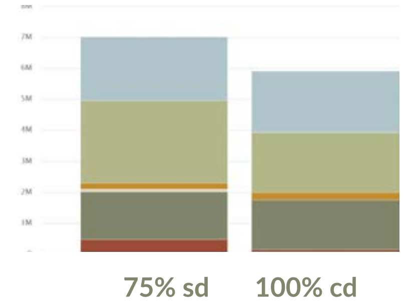
arch 501, fall 2025

gallery perspective view

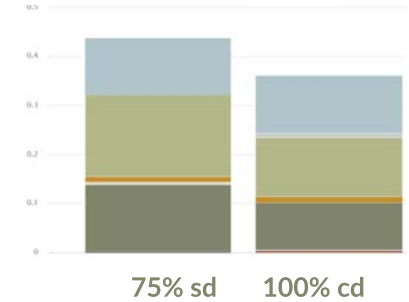


watering plants: a temporal study of the species in my apartment + their movement

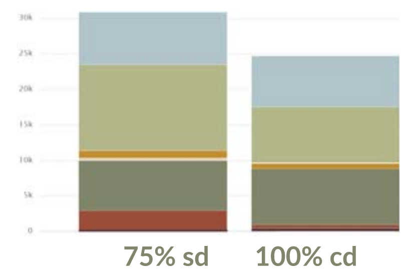
arch 521, fall 2025



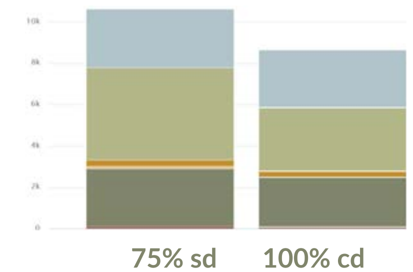
global warming: kg CO2e



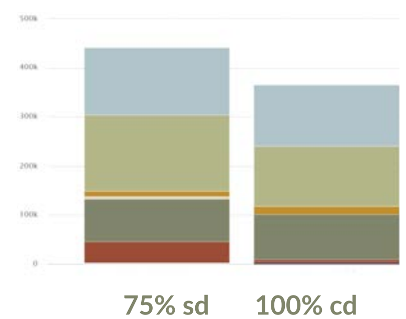
ozone depletion: kg SO2e



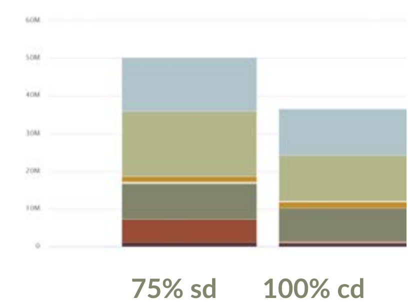
acidification: kg CF11e



eutrophication: kg Ne



tropospheric ozone: kg O3e



non-renewable energy use

what categories are here?

the six impact categories at left are ways our buildings contribute to the atmosphere/environment. the metrics are shown below each graph. there are more impact categories we can measure, LEED recognizes these six though.

LEED v4.1 M+R Credit: WB Life Cycle Assessment

the six graphs above show life cycle impact reduction by category (embodied carbon, ozone depletion, and so on) from the 75% SD to the 100% CD model.

ehdd usually contracts out all life cycle assessment work to a LEED consultant. in this first in-house life cycle study, I explored new ways of visualizing data. rendering + revit model in collaboration with ehdd, data + diagrams all my own work.

- foundation elements
- structural framing/slabs
- sheathing + flashing
- stairs
- exterior walls
- exterior windows

above grade framing + slabs (superstructure)

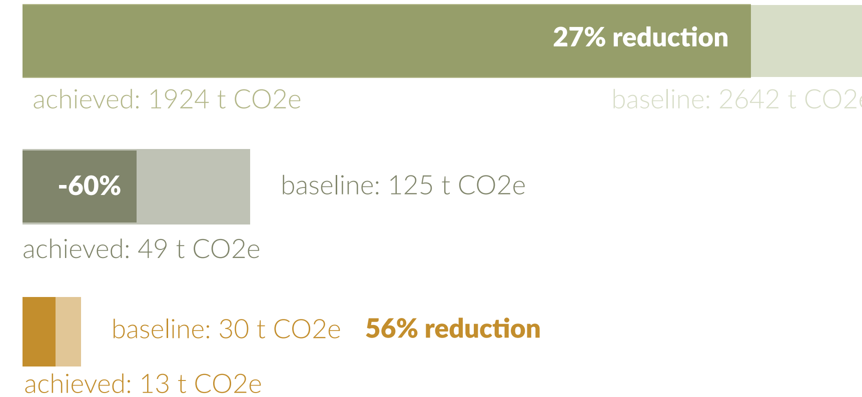
reduction of floor-to-floor height

walls (rainscreen, shear walls, EIFS)

aluminum sheet changed to thin profile

roofing (membrane, insulation)

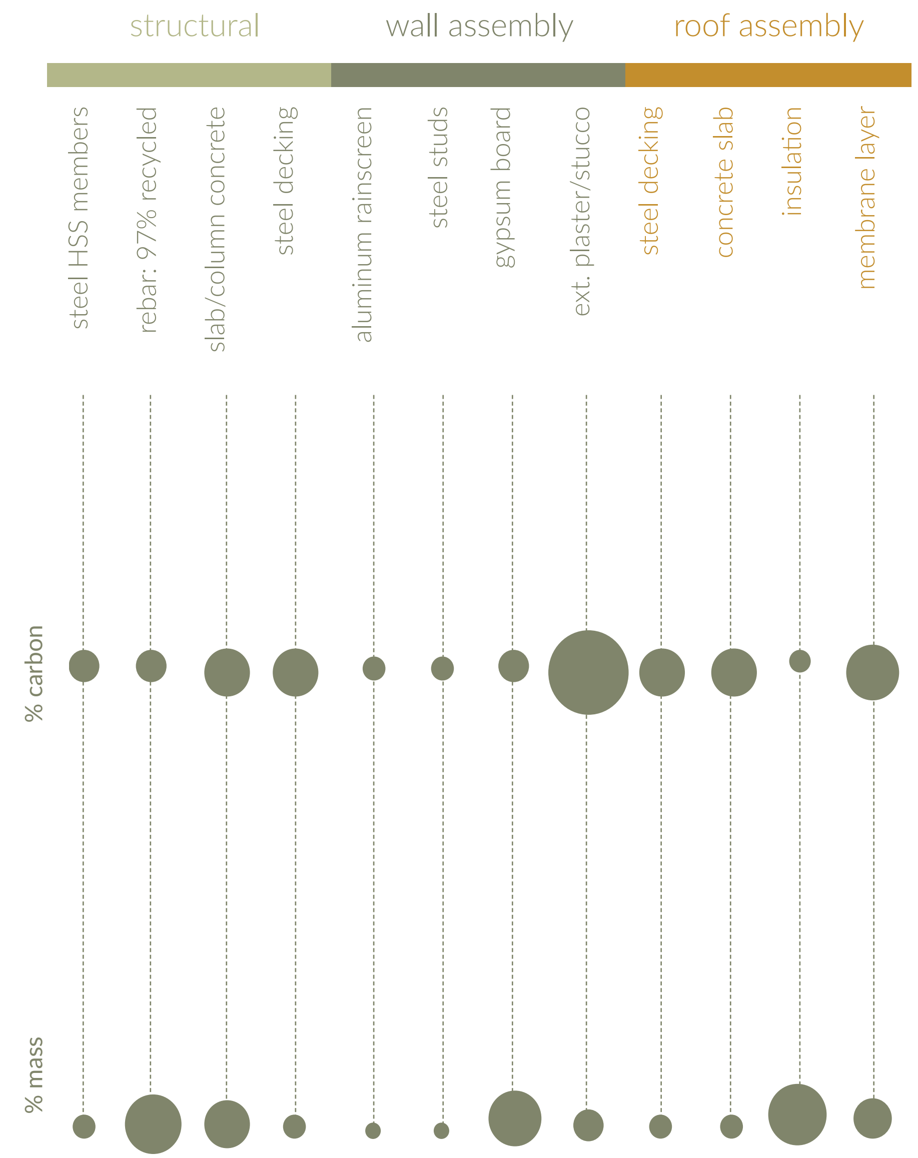
asphalt to modified bituminous membrane



where can we cut carbon?

the graph at left is an alternate visualization of the largest reductions in embodied carbon (t CO2e) by building component.

material impact (% of building kgCO2e) vs weight

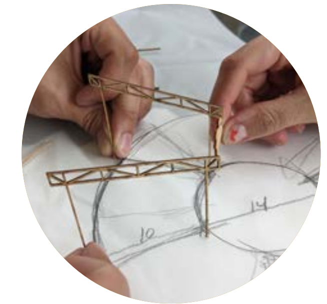


and by which materials?

the graph at left vizualizes, in abstract fashion, the percent embodied carbon of each material compared to its percent mass in the building.

sfsu housing lca

ehdd + carbon leadership forum



step 1. stick model

step 2. truss prefabrication

step 3. welding columns

step 4. panel prefabrication

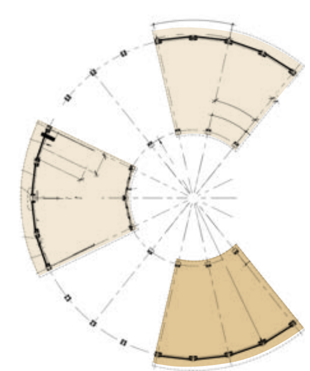
step 5. sitework

step 6. chunk assembly

step 7. connecting chunks

step 8. gravel + watering it in

step 9. celebration

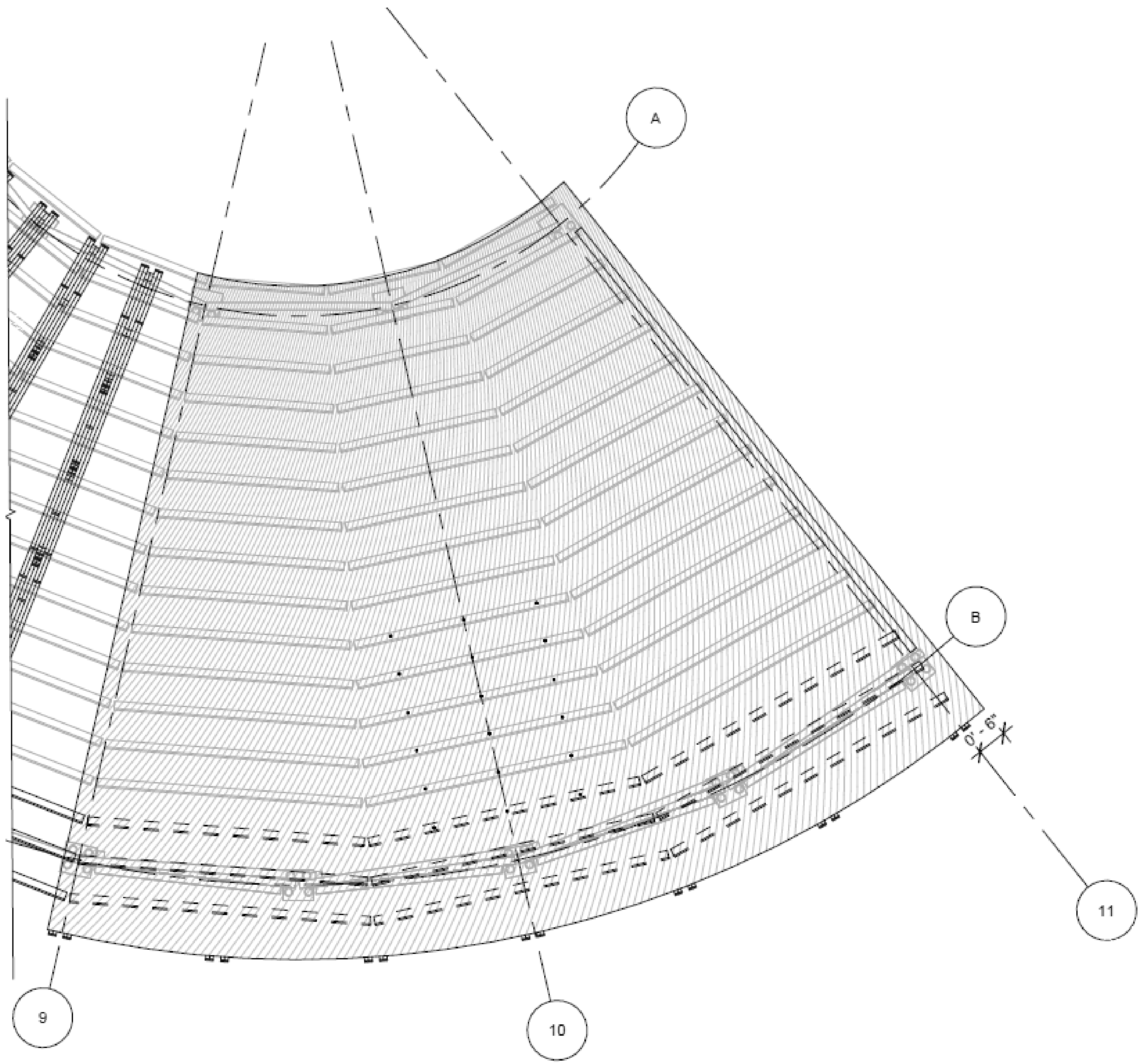


program + the uw farm:

the campus urban farm at the univ. of washington requested our studio build a gathering space for classes and cooking as their programming expanded.

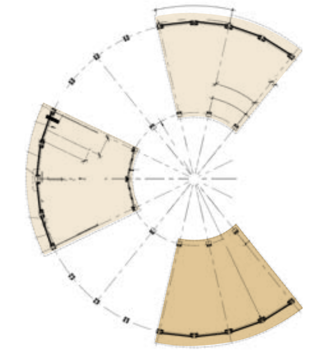
credits for studio work:

team of 16 students and 3 profs led by Prof. Steve Badanes (pictured in step 9). my role was initial sitework coordination with the uw farm and roof detailing.



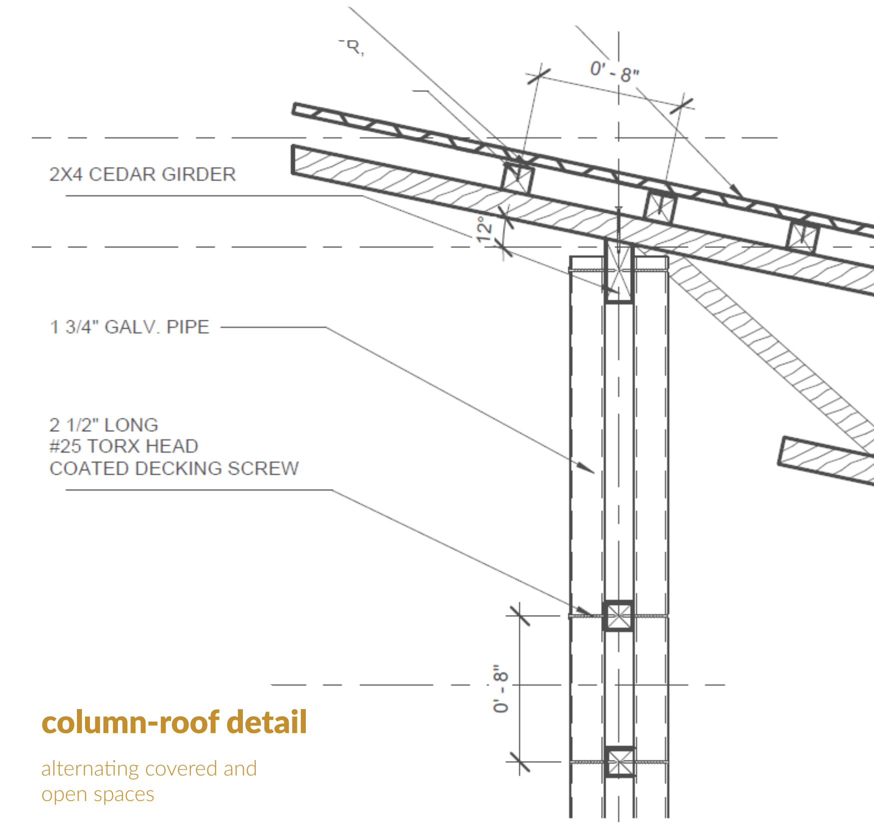
roof "segment" enlarged plan

three 120-sf wedges
two "trellis" spaces



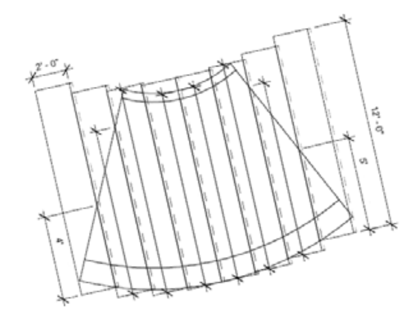
diagrammatic roof plan

alternating covered and
open spaces



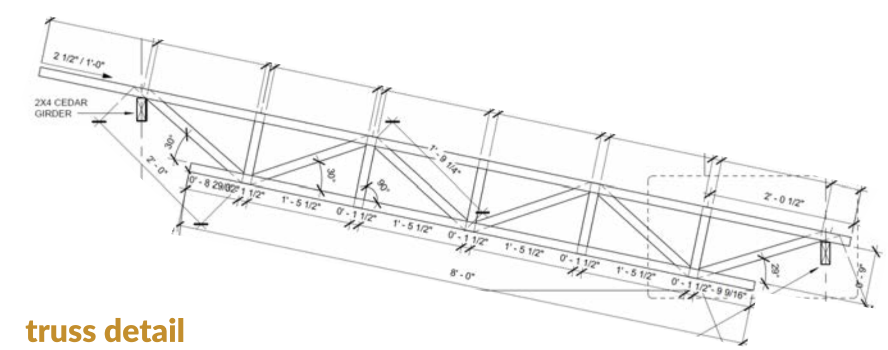
column-roof detail

alternating covered and
open spaces



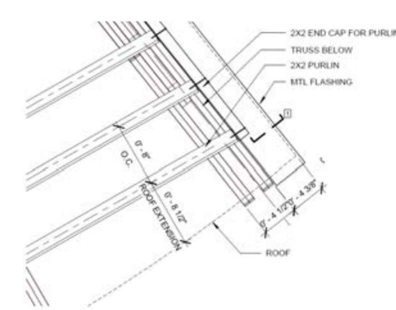
roof panel layout: nts

a way to tile 2' x 12' polycarbonate
panels for each segment



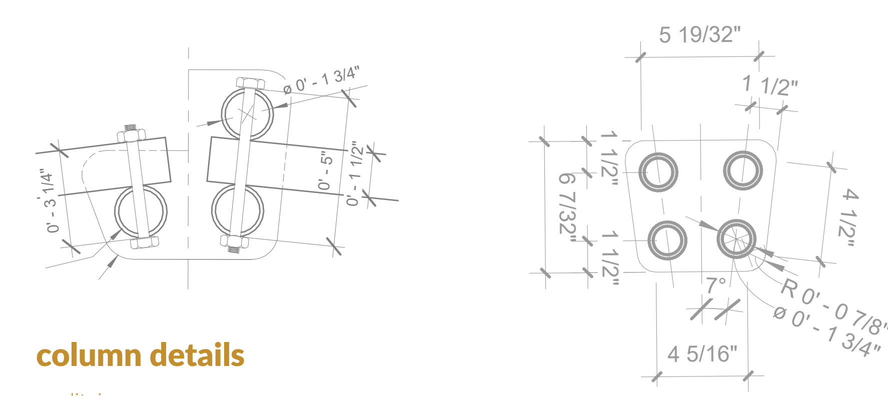
truss detail

credit: kana takagi



roof edge detail: nts

credit: caitlin truong, taylor sanville



column details

credit: ian green

uw farm all details and drawings made using Revit.

uw arch 402, spring 2023

elements

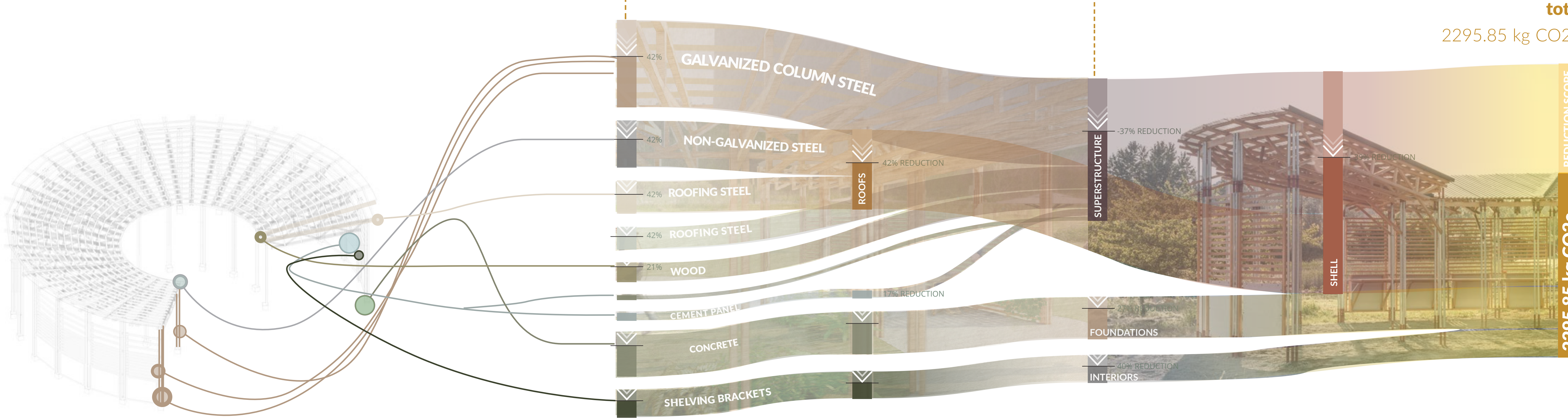
to stay within the budget we used chain-link-fence posts for columns. Exterior columns were comprised of four posts and interior columns two, welded together with plasma cut plates.

categories

the metal superstructure came out to roughly half the total carbon impact. poured concrete footings and stainless steel shelving brackets/fasteners were the rest- with potential for a 37% reduction.

total

2295.85 kg CO_2e.



life cycle assessment 2:

all materials were procured locally from around the Seattle area, and as this was a design-build, we had granular control over what materials we were purchasing.

this study was done retroactively to see if we could have lowered emmissions further than we did. The galvanized column steel was the largest contributor to energy, and understandably so, as the structure was wood.

138 propane cylinders

50 trees growing 10 years

8 months straight of driving a car

uw farm rendering in Revit and Lumion, co-author Taylor Sanville

uw arch 402, spring 2023

wind speed/direction

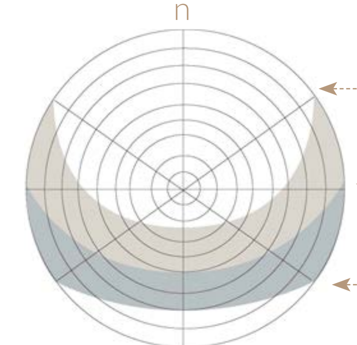
4-7 knots

7-11 knots

11-17 knots

wind speed + massing

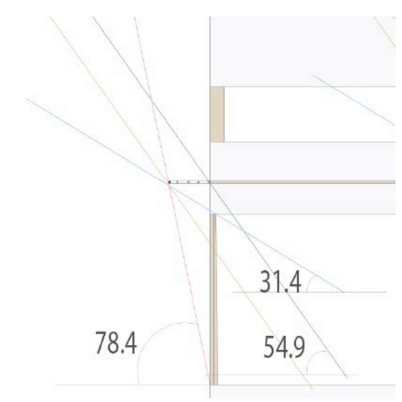
the winds that come primarily from the south and southwest inform our decision to curve the back wall of the adu.



sun paths

sun paths + design

the sun-paths inform the shading device designing shown below.



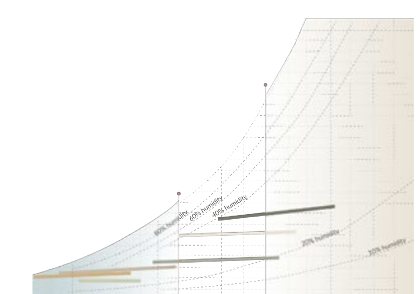
shading device design

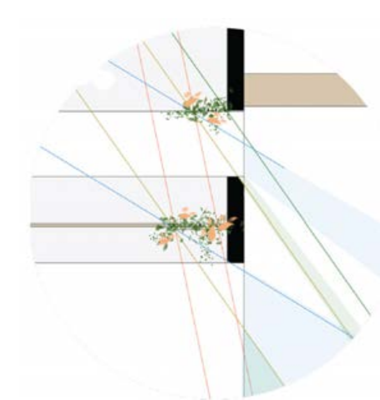
angles of shading trellis based off of sun altitude at solar noon, calculated using the following equation:

A - 90 - latitude + declination

december = 90 - 35.1 - 23.5 = 31.4
june = 90 - 35.1 + 23.5 = 78.4
sept = march = 90 - 35.1 = 54.9

temp/humidity





environmental systems

arch 451 is UW env. systems. an environmentally responsive building begins by mapping the climate - sun, wind, rain, humidity.

adaptive shading: plants

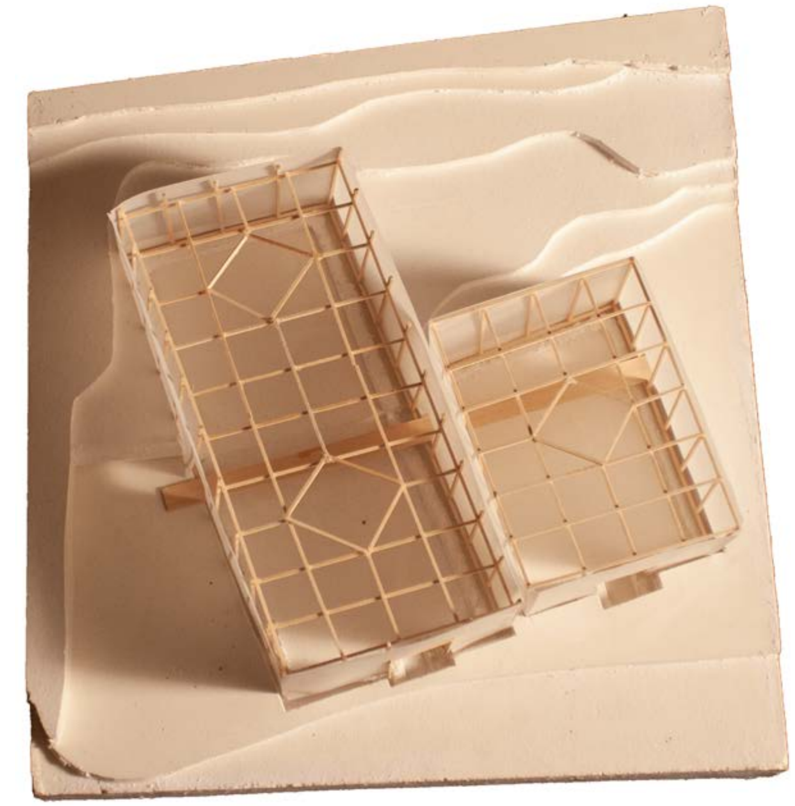
vegetation on a trellis is used to adaptively shade the glazing at the angles calculated above, as shown by diagram top right.

temperature + mass

the day-night temperature and humidity swings and the high temperatures indicated above point to a high-mass building..

plaster base

wooden framework



wooden framework + trace shell (above), paper roofs (below)

credit for studio work:

plaster base formwork: self
plaster pouring: oleh fylyk
slats on outside: jorge burke
wooden framework: self

skeleton and skin

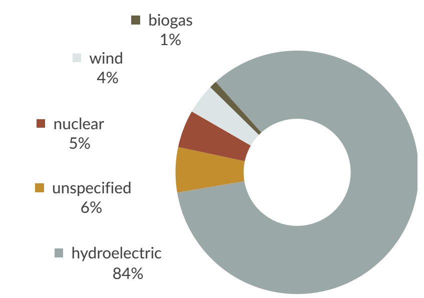
peter zumthor's shelter for roman ruins in chur, switzerland, encapsulates the ruins of a roman structure in a thin wooden skeleton wrapped with a skin of wood slats. A suspended aperture allows in light. a 1/16" massing model and a 1/2" scale section model.

shelter for roman ruins: a tectonic precedent study in plaster, wood, and paper

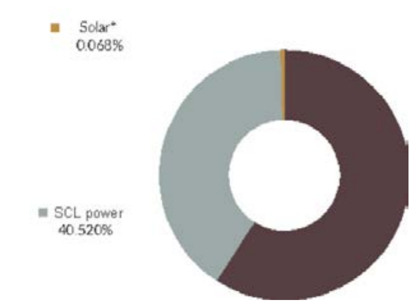
uw arch 300, fall 2021

energy consumption

obtaining accurate energy data for every building on campus was difficult as many still run on pneumatic controls. to counter this, we checked data for each building against others of the same age. energy consumption graphed at left.



seattle city light power mix

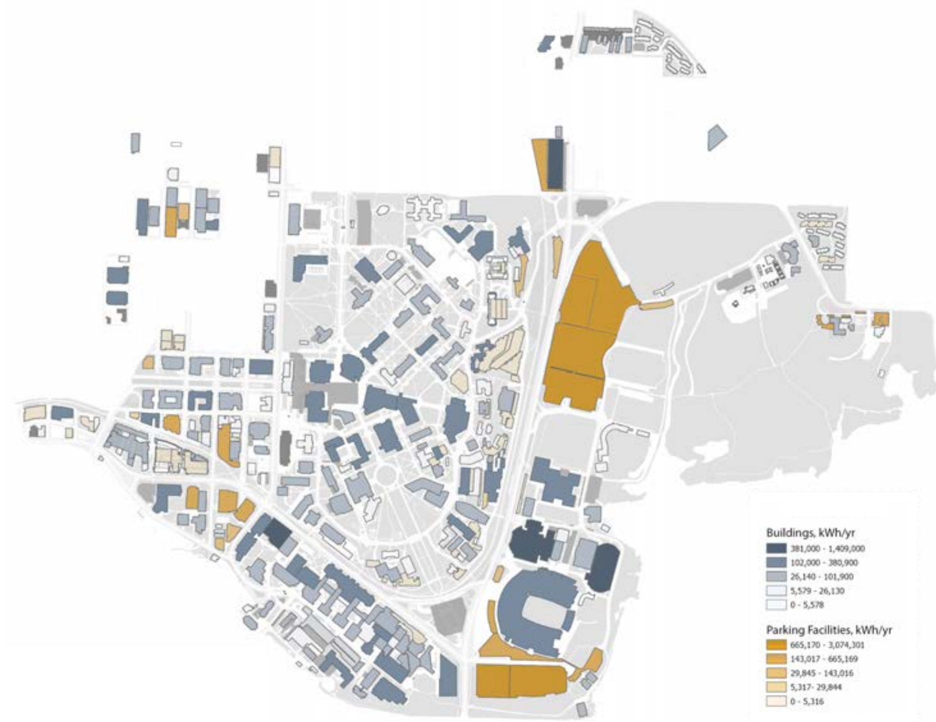


u. of washington power mix

the University's grid is heavily reliant on fossil fuels: transmission infrastructure caps the amount of electricity purchased from the city at roughly 40% of current power use (which will decrease as the university grows and demand increases further). the remainder of power is generated by the on-campus natural gas power and steam plant. Solar can offset this.

campus solar plan: a comprehensive solarization plan made in GIS and Tableau.

uw urbdp 498, fall 2021



pv generation potential

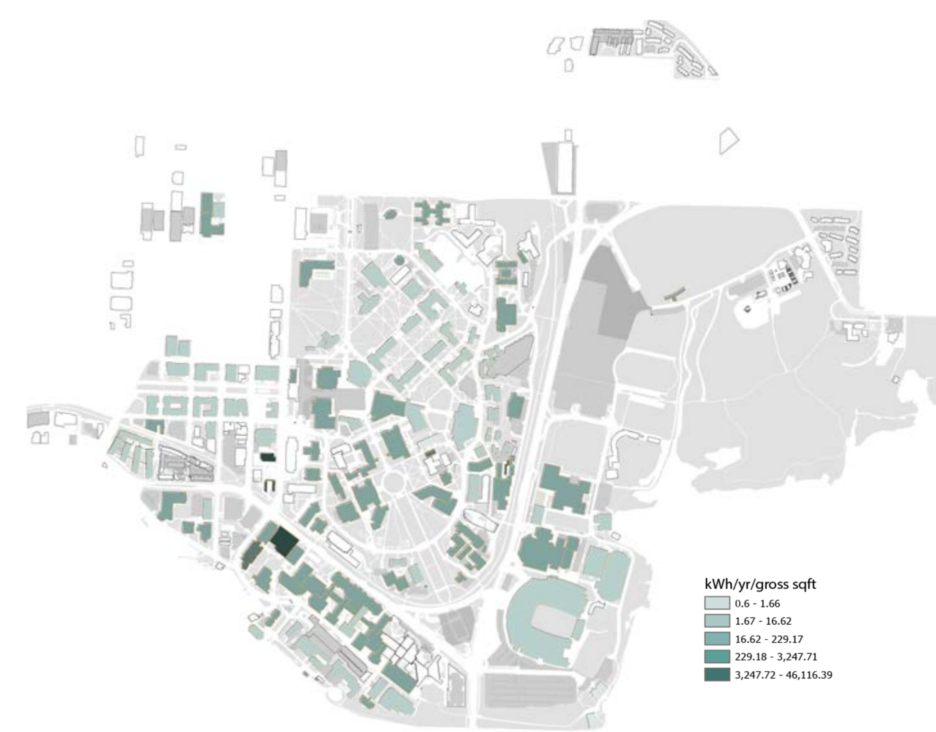
the generation potential for each building is, mainly, a factor of total usable roof area and of shading. NREL's PVWatts system assissted in providing the data in this graph. note the higher generation numbers for parking lots - this is due to sheer area.

solar offset

solar generation potential relative to energy consumption is the solar offset. for buildings, where there are multiple stories of high consumption use, solar typically offsets less than 20% of full building energy use.

solar ranking

a factor of building roof age, date of completion, shading, historic designation, and roof area, the solar ranking of each building is shown at left. "A" buildings are ones on which construction of an array can begin immediately. "C" and "D" buildings are worse.



energy consumption

map intentionally re-iterated from previous page for emphasis.

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